

q -Notes

Ross Street

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1 Review of some q concepts

1. The n -th q -integer

$$[n]_q = \frac{q^n - 1}{q - 1} = 1 + q + q^2 + q^3 + \cdots + q^{n-1}$$

is the cardinality of projective space of dimension n over the field $\mathbb{F}_q$ of cardinality q . That is, it is the number of one-dimensional subspaces of $\mathbb{F}_q$. Note that $[n]_1 = n$.

2. The q -factorial

$$[n]_q! = [n]_q [n-1]_q \cdots [2]_q [1]_q$$

is the number of complete flags $0 < V_1 < V_2 < \cdots < V_n = V$ in an n -dimensional $\mathbb{F}_q$ -vector space V .

3. Put

$$\phi_n(q) = (q^n - 1)(q^{n-1} - 1) \cdots (q - 1) = (q - 1)^n [n]_q! .$$

If V is an n -dimensional $\mathbb{F}_q$ -vector space then the number of linear bijections from V to itself is

$$\#\mathrm{GL}(n, q) = (q^n - 1)(q^n - q) \cdots (q^n - q^{n-1}) = q^{\frac{1}{2}n(n-1)} \phi_n(q) .$$

4. The q -binomial coefficient

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]_q!}{[k]_q! [n-k]_q!} = \frac{\phi_n(q)}{\phi_k(q) \phi_{n-k}(q)}$$

is the number of k -dimensional subspaces of an n -dimensional $\mathbb{F}_q$ -vector space.

5. The q -derivative of a function $f(x)$ is defined by

$$(D_q f)(x) = \frac{f(qx) - f(x)}{qx - x} .$$

6. The q -integral of a function $f(x)$ is defined by

$$(I_q f)(x) = \sum_{k \geq 0} (1 - q)xq^k f(q^k x) .$$

These definitions make sense and can be applied even when q is not a prime power.

Example 1.1.

$$(D_q f(a-))(x) = \frac{f(qax) - f(ax)}{qx - x} = a \frac{f(qax) - f(ax)}{qax - ax} = a(D_q f)(ax) .$$

Example 1.2. Suppose $f(x) = x^n$. Then

$$(D_q f)(x) = \frac{(qx)^n - x^n}{qx - x} = \frac{(q^n - 1)x^n}{(q - 1)x} = [n]_q x^{n-1} .$$

Example 1.3. $D_q(f + g) = D_q f + D_q g$.

Let us write $f \cdot g$ for the pointwise product $(f \cdot g)(x) = f(x)g(x)$ of functions. We have the q -Leibniz formula:

Example 1.4. $D_q(f \cdot g) = D_q f \cdot g(q-) + f \cdot D_q g = D_q f \cdot g + f(q-) \cdot D_q g$.

Proposition 1.5. $D_q I_q f = f$.

Proof. Using these examples, we can first calculate

$$\begin{aligned} (D_q(-) \cdot f(q^k-))(x) &= qx \cdot (D_q f(q^k-))(x) + (D_q(-))(x) \cdot f(q^k x) \\ &= qxq^k (D_q f)(q^k x) + f(q^k x) \\ &= xq^{k+1} \frac{f(q^{k+1}x) - f(q^k x)}{q^{k+1}x - q^k x} + f(q^k x) \\ &= \frac{q}{q-1} f(q^{k+1}x) - \frac{1}{q-1} f(q^k x) . \end{aligned}$$

Hence by multiplying by $(1 - q)q^k$ and summing, we see that

$$\begin{aligned} (D_q I_q f)(x) &= \sum_{k \geq 0} (q^k f(q^k x) - q^{k+1} f(q^{k+1} x)) \\ &= f(x) - qf(qx) + qf(qx) - q^2 f(q^2 x) + q^2 f(q^2 x) - q^3 f(q^3 x) + \dots \\ &= f(x) \end{aligned}$$

as all but the first term cancel. □

The q -exponential function is defined by

$$e_q(x) = \sum_{n \geq 0} \frac{x^n}{[n]_q!} = \sum_{n \geq 0} (q-1)^n \frac{x^n}{\phi_n(q)} .$$

Using Examples 1.1, 1.2 and (the infinite version of) 1.3, we have that the q -exponential is fixed under q -differentiation:

$$D_q e_q = e_q . \quad (1.1)$$

Using Example 1.4, we have an antiderivative formula:

Proposition 1.6. *Suppose $D_q f_n = f_{n+1}$ and $D_q g_{n+1} = g_n(q-)$ for all $n \geq 0$. Then*

$$f_0 \cdot g_0(q-) = D_q \sum_{k \geq 0} (-1)^k f_k \cdot g_{k+1} .$$

Apply this to $f_n(x) = (D_q^n f)(x)$ and $g_0(x) = 1$, $g_n(x) = q^{\frac{1}{2}n(n-1)} \frac{x^n}{[n]_q!}$ to obtain the following formula for a q -antiderivative for any function f .

$$\int_q f(x) dx = \sum_{n \geq 0} (-1)^n q^{\frac{1}{2}n(n+1)} (D_q^n f)(x) \frac{x^{n+1}}{[n+1]_q!} \quad (1.2)$$

From Example 1.2, we deduce:

Proposition 1.7. *The q -derivative of*

$$f(x) = \sum_{n \geq 0} a_n \frac{x^n}{[n]_q!}$$

is

$$(D_q f)(x) = \sum_{n \geq 0} a_{n+1} \frac{x^n}{[n]_q!} .$$

Let us now make a connection with our paper [4] in which we studied the q -general linear groupoid $\mathcal{GL}_q$ of finite $\mathbb{F}_q$ -vector spaces with linear bijections. The external tensor product $M \otimes N$ of $[\mathcal{GL}_q, \text{Vect}_{\mathbb{C}}]$ was defined in Section 2 of [4] by

$$(M \otimes N)(A) = \bigoplus_{C \leq A} M(C) \otimes N(A/C) . \quad (1.3)$$

In Section 3, we looked at the *dimension series*

$$(\dim M)(x) = \sum_{n \geq 0} \dim M(\mathbb{F}_q^n) \frac{x^n}{\phi_n(q)} \quad (1.4)$$

of functors $M : \mathcal{GL}_q \rightarrow \text{Vect}_{\mathbb{C}}$ landing in finite dimensional complex vector spaces. In Section 3 of [4], it was shown (involving the q -binomial coefficient) that

$$\dim(M \otimes N) = \dim M \cdot \dim N . \quad (1.5)$$

Now we would like to modify this a little and define a q -Taylor dimension series by

$$(\text{dit} M)(x) = \sum_{n \geq 0} \dim M(\mathbb{F}_q^n) \frac{x^n}{[n]_q!} = (\dim M)((q-1)x) \quad (1.6)$$

We also have

$$\text{dit}(M \otimes N) = \text{dit} M \cdot \text{dit} N . \quad (1.7)$$

Encouraged by Proposition 1.7, we define the derivative functor $M' : \mathcal{GL}_q \rightarrow \text{Vect}_{\mathbb{C}}$ of M by

$$M'(A) = M(A \oplus \mathbb{F}_q) \quad (1.8)$$

to obtain

$$\text{dit} M' = D_q \text{dit} M . \quad (1.9)$$

Let us look at the derivative of $M \otimes N$:

$$\begin{aligned} (M \otimes N)'(A) &= \bigoplus_{C \leq A \oplus \mathbb{F}_q} M(C) \otimes N(A \oplus \mathbb{F}_q / C) \\ &\cong \bigoplus_{C \leq A} M(C) \otimes N(A/C \oplus \mathbb{F}_q) \oplus \bigoplus_{C \leq A} M(C \oplus \mathbb{F}_q) \otimes N(A/C) \\ &\cong (M \otimes N')(A) \oplus (M' \otimes N)(A) \end{aligned}$$

since the subspaces of $A \oplus \mathbb{F}_q$ are either of the form C or $C \oplus \mathbb{F}_q$ for $C \leq A$, but not both. So it seems we have the Leibniz-like formula

$$(M \otimes N)' \cong M \otimes N' \oplus M' \otimes N . \quad (1.10)$$

Wait a minute! These formulas do not seem consistent with Example 1.4. For we have the calculation

$$\begin{aligned} \text{dit} M \cdot D_q \text{dit} N + D_q \text{dit} M \cdot \text{dit} N &\stackrel{1.9}{=} \text{dit} M \cdot \text{dit} N' + \text{dit} M' \cdot \text{dit} N \\ &\stackrel{1.7}{=} \text{dit}(M \otimes N' \oplus M' \otimes N) \\ &\stackrel{1.10}{=} \text{dit}(M \otimes N)' \\ &\stackrel{1.9}{=} D_q \text{dit}(M \otimes N) \\ &\stackrel{1.7}{=} D_q(\text{dit} M \cdot \text{dit} N) \\ &\stackrel{\text{Ex 1.4}}{=} (\text{dit} M)(q-) \cdot D_q \text{dit} N + D_q \text{dit} M \cdot \text{dit} N \end{aligned}$$

which looks false since $(\text{dit} M)(x) \neq (\text{dit} M)(qx)$. Where is the mistake?

References

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